

Graph-Based Deep Learning for Multi-Class Eye Disease Classification Using Deep Convolutional Features

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Abstract

Automated analysis of retinal fundus images can assist the screening of several ocular disorders, but visually similar manifestations and substantial differences in class frequency make multi-class classification difficult. This study investigates a graph-based learning pipeline in which a pretrained ResNet50 first converts each 224×224 fundus image into a spatial feature map and a fixed two-dimensional grid is then used to represent the feature map as a graph. Four graph classifiers—MDGNet, a Visual Graph Network (ViG), GATv2, and SuperGAT—are evaluated against the ResNet50 classifier used as the convolutional baseline. The experiments use 5335 images from ten eye-disease categories. A stratified 80/20 split produces 4268 training images and 1067 held-out test images; the training portion is further divided into 3414 graph-training images and 854 validation images. Early stopping is based only on validation performance, while the final test set remains untouched during model selection. Among the evaluated graph models, MDGNet obtains the highest test accuracy (76.76%), whereas the ResNet50 baseline produces the highest macro one-vs-rest AUC (0.9578). SuperGAT reaches 76.48% accuracy, while the five-fold ViG probability ensemble reaches 75.07%. The results indicate that converting convolutional feature maps into spatial graphs is feasible, but the gains over the strong CNN baseline are modest and depend on the graph architecture.

Keywords: eye disease classification; fundus imaging; graph neural network; ResNet50; MDGNet; Visual GNN; GATv2; SuperGAT; deep learning; medical image analysis

BDEyeMS-GCN: Multi-Scale Graph Convolutional Network for Multi-Class Ocular Pathology Recognition from Bangladeshi Clinical Fundus Images

ABSTRACT

Multi-class ocular pathology recognition from fundus images remains a critical clinical challenge, particularly in densely populated regions like Bangladesh where expert ophthalmological resources are scarce. Traditional diagnostic workflows rely heavily on manual feature extraction, which is susceptible to inter-observer variability. While Convolutional Neural Networks (CNNs) have advanced automated screening, they fundamentally process images as Euclidean grid structures, failing to adequately capture the complex topological relationships and spatial distribution patterns of scattered subtle lesions (e.g., microaneurysms, hemorrhages, and exudates). To overcome these architectural limitations, we propose the BDEyeMS-GCN (Multi-Scale Graph Convolutional Network), an automated framework optimized for Bangladeshi clinical fundus images. This architecture isolates pathological regions across varying scales and constructs a topological graph to model the spatial dependencies of lesion nodes. By integrating multi-scale local feature extraction with global graph convolutions, BDEyeMS-GCN dynamically assigns weight to highly

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variant ocular morphologies. This structural combination minimizes information loss in subtle lesion detection and provides a robust, scalable diagnostic tool for high-burden clinical environments.

INDEX TERMS

Diabetic Retinopathy, Fundus Images, Graph Convolutional Networks, Multi-Class Ocular Pathology, Multi-Scale Feature Extraction, Teleophthalmology.

1. INTRODUCTION

Preventable vision impairment and irreversible blindness pose serious clinical and public health challenges globally, with a significant disease burden in low- and middle-income regions such as Bangladesh. Pathologies including Diabetic Retinopathy (DR), Glaucoma, Age-Related Macular Degeneration (AMD), hypertensive retinopathy, and cataract-induced fundus alterations require timely detection to prevent permanent loss of sight. Color Fundus Photography (CFP) serves as a primary, non-invasive diagnostic modality in ophthalmology. However, population-scale manual assessment of fundus imagery by expert ophthalmologists is hampered by a shortage of specialists, inter-observer diagnostic variability, and high patient volume in rural and regional screening centers.

Recent developments in Artificial Intelligence (AI) and Machine Learning (ML), specifically Computer-Assisted Diagnostic (CAD) frameworks based on Deep Learning (DL), have enabled automated medical image interpretation. While Deep Convolutional Neural Networks (CNNs) excel at extracting localized translation-invariant patterns from grid-structured pixel arrays, they exhibit structural constraints when capturing non-local relationships, morphological variations, and spatial distribution patterns across distant retinal regions. Graph Convolutional Networks (GCNs) address these limitations by modeling arbitrary geometric relationships, representing localized image regions or lesions as graph nodes and their topological interactions as weighted edges.

A. PROBLEM STATEMENT

Ocular pathologies, including diabetic retinopathy (DR) and glaucoma, represent a leading cause of irreversible visual impairment worldwide. In Bangladesh, the rising incidence of diabetes mellitus has precipitated a proportional increase in DR and related retinal vascular disorders. Detecting these pathologies requires the precise identification of microvascular alterations, such as hard exudates, microaneurysms, and intraretinal hemorrhages. Manual evaluation of fundoscopic images by clinical specialists is labor-intensive and creates substantial diagnostic bottlenecks in under-resourced healthcare systems.

Although machine learning algorithms, specifically Convolutional Neural Networks (CNNs), provide automated diagnostic pathways, their performance degrades when evaluating complex, overlapping pathologies. CNNs excel at extracting localized grid-based features but struggle to map the non-Euclidean geometric relationships between disparate retinal structures, such as the spatial correlation between the optic disc and peripheral neovascularization. Furthermore, standard CNNs apply uniform processing scales, rendering them ineffective at distinguishing minute, subtle lesions from background noise. Developing an automated system capable of recognizing these multifaceted spatial dependencies is essential for accurate, multi-class ocular disease grading in diverse, real-world clinical settings.

B. OBJECTIVES

Architectural Integration: Design a hybrid architecture (BDEyeMS-GCN) that integrates multi-scale convolutional operations with graph-based representations to capture both localized lesion morphology and global topological contexts.

Multi-Scale Feature Fusion: Implement a dynamic fusion module capable of extracting features across varying lesion sizes—from pinpoint microaneurysms to expansive macular edema—and automatically assigning fusion weights based on diagnostic contribution.

Clinical Validation: Assess the robustness and generalization of the model specifically on Bangladeshi clinical fundus datasets, establishing a scalable screening mechanism tailored for under-resourced environments.

C. OVERVIEW

This paper introduces the BDEyeMS-GCN framework for multi-class ocular pathology classification. Section II systematically reviews the related literature concerning deep learning and graph-based methodologies in ophthalmic imaging. Section III details the methodology, defining the multi-scale feature extraction protocol and the graph construction mechanics. Section IV presents the experimental setup, dataset characteristics, and quantitative evaluation against baseline architectures. Finally, Section V concludes the research and delineates future trajectories for clinical deployment.

2. RELATED WORK

Recent advancements in deep learning have catalyzed the automation of retinal disease diagnostics. However, transitioning from purely localized feature extraction to holistic structural representation remains an active area of research. This section synthesizes the current literature, progressing from baseline convolutional networks to advanced graph-based and multi-modal architectures.

Early automated grading models heavily utilized CNNs to isolate ocular lesions. For instance, Wang et al. [?] proposed Zoom-in-Net to deeply mine lesion features, mimicking the diagnostic process of clinicians zooming into high-resolution fundus areas. Li et al. [?] expanded upon this by introducing CANet, a cross-disease attention network capable of jointly grading DR and diabetic macular edema (DME). To further refine the isolation of subtle structural abnormalities, Kulkarni and Reddy [?] recently developed the OGFA+CNN approach, integrating custom graph filters as an augmentation step before standard CNN processing to enhance complex blood vessel relationships.

Despite their utility, CNNs process images as standard grids, ignoring topological structures. To resolve this, researchers have turned to GNNs. Sundar and Sumathy [?] demonstrated that extracting topological features via Variational Autoencoders (VAEs) and mapping them through GCNs significantly improves the identification of complex severity levels. Zhang et al. [?] proposed a Hybrid Graph Convolutional Network (HGCN) that fuses CNN grid features with GCN topological features, while Liu et al. [?] introduced GREEN, a graph residual re-ranking network that models the relationships between disease severity levels in the feature space.

Lesion scale variance requires dynamic network adaptability. Wang et al. [?] developed MDGNet, which utilizes dynamic weight fusion on multi-scale local features, automatically learning the diagnostic contribution of different lesion sizes. Addressing the spatial distribution of these lesions, Zhang et al. [?] proposed SIGraph, employing saliency perception to convert critical pixel regions into lesion graph nodes. Similarly, Qian et al. [?] structured the Pyramid Vision Graph Convolutional Network (PVGCN) for Optical Coherence Tomography (OCT), decomposing images into sub-images across scales to replace heavy dimensional extraction with lightweight graph pooling. Poranki and Rao [?] integrated the Synthetic Minority Over-sampling Technique (SMOTE) with a Deep Graph Correlation Network (DRG-Net) and XGBoost classifier to handle severe data imbalance during grading.

The prohibitive cost of manual annotation has driven the development of semi-supervised and transfer learning models. Duan et al. [?] successfully combined CNNs with attentional GCNs in a semi-supervised framework, substantially improving performance

with limited labels. Zhang et al. [?] completely removed the need for manual annotations by using a Deep Graph Correlation Network (DGCN) to cluster unlabelled retinal images based on inherent feature correlations. To tackle inter-domain camera variations, Hu et al. [?] designed Graph Adversarial Transfer Learning (GATL), aligning features across different datasets. Finally, addressing the necessity of holistic patient evaluation, Zedadra et al. developed VisionTrack [?], a multi-modal approach synthesizing OCT, fundus images, and clinical text via Large Language Models (LLMs), followed by a Graph-Aware Multimodal architecture [?] to predict multi-label retinal diseases. [?] Yu and Pei proposed a Dynamic Graph Clustering Learning (DGCL) framework for unsupervised diabetic retinopathy classification by combining CNN and GCN features with dynamic graph clustering. Their method eliminates the need for manual annotations while improving clustering stability and classification performance through multi-structural feature fusion and dynamic memory updating.

[?] Lenka et al. introduced a graph convolution-based multi-task model for glaucoma detection that simultaneously performs optic disc and optic cup segmentation and disease classification. The proposed MobileNetV2-U-Net with residual GCN and attention modules achieved 97.43

[?] Akter et al. presented a graph-enhanced deep learning framework for diabetic retinopathy diagnosis by integrating pre-trained feature extractors with Graph Convolutional Networks, quality assessment, and uncertainty estimation. Their approach achieved state-of-the-art performance on APTOS2019, Messidor-2, and EyePACS datasets while improving prediction reliability and interpretability.

[?] Lux et al. proposed a biology-informed heterogeneous graph representation for OCTA images to provide interpretable diabetic retinopathy staging. Their graph neural network models retinal vessels, intercapillary areas, and the foveal avascular zone, enabling accurate disease prediction while offering clinically meaningful explanations of retinal abnormalities.

TABLE 1. Summary of Related Work

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Year | Author(s) | Findings | Contribution | Research Gap 2017 | Z. Wang et al.[?] | Jointly localizing lesions and classifying diabetic retinopathy (DR) significantly improves grading accuracy compared to whole-image classification alone. | Introduced Zoom-in-Net, a multi-stage framework that automatically localizes lesion-attentive regions and fuses local lesion features with global image context. | Relies heavily on bounding box-like local cropping; fails to capture complex topological or non-Euclidean spatial relationships among distant lesions. 2019 | X. Li et al.[?] | Jointly modeling DR and Diabetic Macular Edema (DME) improves feature representations via cross-disease attention. | Proposed CANet, a dual-branch attention network that models inter-disease dependencies and feature correlation across DR and DME. | Focuses primarily on grid-structured CNN feature maps, limiting its ability to model irregular geometric structures and lesion graph networks. 2025 | P. Kulkarni K. S. Reddy[?] | Combining meta-heuristic optimization with CNNs enhances multi-level ocular disease feature selection and classification performance. | Developed an OGFA+CNN approach integrating Opposition-based Grasshopper Optimization (OGFA) for optimal hyperparameter selection and feature mining in fundus images. | Lacks structural graph modeling of lesions, which restricts its capability to capture topological relationships across retinal biomarkers. 2023 | S. Sundar S. Sumathy [?] | Graph Neural Networks (GNNs) effectively capture topological relationships and spatial patterns of retinal lesions for DR grading. | Proposed a GNN framework that constructs retinal topology graphs from fundus images to extract structural disease features. | Highly sensitive to graph construction methods; performance degrades when noise or image artifacts disturb node feature extraction. 2021 | G. Zhang et al. [?] | Graph Convolutional Networks (GCNs) effectively propagate label information to leverage unlabelled retinal images in semi-supervised paradigms. | Formulated a Hybrid GCN combining local image features with global sample-relation graphs for semi-supervised retinal image classification. | Sample-level relation graphs scale poorly with massive datasets and struggle when inter-class boundaries in unlabelled data are highly ambiguous. 2020 | S. Liu et al. [?] | Pairwise relationship modeling among candidate fundus images via graph re-ranking significantly improves fine-grained DR grading. | Developed GREEN, a graph residual re-ranking network that refines feature embeddings by modeling structural similarities between patient fundus scans. | Re-ranking requires comparison with reference sample sets, increasing computational latency during inference on large-scale datasets. 2024 | Y. Wang et al. [?] | Multi-scale local feature aggregation combined with dynamic graph weighting accurately captures varying lesion sizes. | Designed a multi-scale GCN architecture incorporating dynamic weight fusion to adaptively aggregate local retinal features. | Dynamic weighting increases computational complexity and remains sensitive to fine-grained background noise in low-quality fundus scans. 2025 | P. Zhang et al. [?] | Combining saliency map awareness with spatial graph aggregation improves detection of subtle, scattered retinal lesions. | Proposed SIGraph, introducing saliency perception blocks paired with edge-cut and node-cluster graph pooling for spatial pattern learning. | Saliency extraction can be hindered by severe illumination variations or non-pathological retinal artifacts (e.g., optic disc glare). 2025 | J. Qian et al. [?] | Vision GCNs with pyramid structures alleviate over-smoothing and capture hierarchical layer representations in OCT images. | Developed PVGCN, combining a Vision Graph block (Grapher + FFN) with a multi-scale pyramid structure for parameter-efficient OCT classification. | Graph construction based on patch connectivity lacks explicit biological semantic mapping (e.g., individual retinal layer boundaries). 2024 | V. K. R. Poranki B. S. Rao [?] | Graph-based representation learning integrated with XGBoost classifiers yields robust DR grading on tabular/extracted graph features. | Proposed DRG-Net, combining graph structural representation learning with an Extreme Gradient Boosting classifier for

final DR severity grading. | The pipeline relies on a multi-stage disconnected architecture, preventing end-to-end gradient optimization from pixels to classification. 2022 | S. Duan et al. [?] | Fusing spatial CNN feature maps with GCN topology reduces reliance on large annotated fundus datasets. | Proposed a semi-supervised hybrid CNN-GCN architecture that learns localized visual features alongside inter-image structural connections. | Prone to noise propagation across graph edges if unlabelled images possess severe artifacts or poor contrast. 2022 | G. Zhang et al. [?] | Self-supervised dynamic graph correlation networks can learn discriminative DR representations without pixel-level manual annotations. | Created a Deep Graph Correlation Network that models cross-image sample correlations without requiring dense manual lesion segmentations. | Performance heavily relies on the assumption of well-curated image clusters, making it susceptible to domain shifts across different camera brands. 2022 | J. Hu et al. [?] | Graph adversarial domain adaptation reduces cross-dataset domain shifts in DR classification. | Proposed a Graph Adversarial Transfer Learning framework that aligns source and target domain features using structural graph alignment. | Requires access to target domain unlabeled data during training, limiting zero-shot adaptation to completely unseen clinical environments. 2025 | A. Zedadra et al. [?] | Multimodal fusion of OCT (structural depth) and Fundus (surface color) images enhances multi-label retinal disease prediction accuracy. | Developed a hybrid multi-modal AI framework fusing deep features from paired fundus photography and OCT scans. | High sensitivity to spatial misalignments between paired OCT and fundus acquisitions; unusable if one modality is missing. 2025 | A. Zedadra et al. [?] | Graph-based multimodal integration models non-linear structural correlations across modalities better than late vector concatenation. | Proposed a Graph-Aware Multimodal Deep Learning architecture that constructs inter-modality graphs for joint DR classification. | Graph construct overhead grows significantly as resolution increases, requiring high computational resources for real-time deployment. 2023 | C. Yu H. Pei [?] | Dynamic graph clustering enables unannotated DR pattern discovery by dynamically updating node cluster relationships during training. | Formulated a Dynamic Graph Clustering Learning framework for unsupervised DR classification using self-organizing graph topologies. | Unsupervised cluster assignments can align with non-clinical features (e.g., image contrast or background tint) rather than true DR pathology severity. 2025 | S. Lenka et al. [?] | Multi-task graph convolutional learning improves glaucoma indicator detection (optic disc/cup analysis) simultaneously with DR features. | Designed a GCN-based multi-task framework for joint glaucoma detection and structural fundus feature extraction. | Task balancing weights require extensive heuristic tuning, and performance degrades if one task significantly dominates optimization gradients. 2025 | Z. Akter et al. [?] | Incorporating image quality estimation and uncertainty quantification into GCNs prevents false diagnoses on degraded fundus images. | Proposed a quality-aware and uncertainty-driven graph-enhanced deep learning framework for DR diagnosis under variable image quality conditions. | High computational overhead due to uncertainty estimation iterations (e.g., Monte Carlo sampling or ensemble graph passes). 2025 | L. Lux et al. [?] | Abstracting OCTA scans into biological graphs (vessels, FAZ, intercapillary zones) yields clinical interpretability without sacrificing accuracy. | Created a Biology-Informed Heterogeneous Graph representation for OCTA DR staging, converting biological structures directly into graph nodes. | Requires accurate initial segmentation of microvasculature and FAZ boundaries; segmentation errors propagate directly into the graph model.

1. Introduction

Fundus photography provides a direct view of retinal structures that are relevant to the assessment of several ocular disorders. In practice, however, a single image may

contain multiple visual cues, and the distinction between neighboring disease categories can be subtle. This makes automated multi-class classification more demanding than binary screening tasks. Convolutional neural networks (CNNs) have become a standard approach for extracting hierarchical visual representations from medical images, with residual learning providing a particularly effective mechanism for training deep networks [1].

A limitation of treating an image only as a regular tensor is that the spatial relationships between learned local representations are handled implicitly by convolution. Graph neural networks (GNNs) provide another way to model such relationships by defining entities as nodes and their interactions as edges. The graph convolutional formulation of Kipf and Welling [2] and attention-based graph learning [3] established widely used mechanisms for propagating information over graph neighborhoods. Later work such as GATv2 [4] showed that attention scoring can be made more expressive, while SuperGAT introduced explicit supervision of attention patterns [5]. Visual graph networks have also been proposed for applying graph reasoning to visual representations [6].

The present work examines whether these ideas provide a useful complement to a conventional CNN for multi-class fundus-image classification. Rather than constructing a graph directly from raw pixels, the proposed pipeline uses the final convolutional feature map of a pretrained ResNet50. The 7×7 spatial locations are treated as graph nodes, and neighboring locations in the feature map are connected using a four-neighbor grid. Each node therefore retains both deep visual information and its normalized spatial coordinates. This design keeps the graph construction deterministic and makes the comparison between graph architectures straightforward.

The main contributions are threefold. First, a reproducible CNN-to-graph pipeline is implemented for ten-class eye-disease recognition. Second, four graph architectures with different inductive biases are compared under the same extracted features and graph topology. Third, the evaluation reports not only accuracy but also class-wise precision, recall, F1-score, and macro one-vs-rest AUC, together with a held-out test protocol and a five-fold cross-validation experiment for the ViG model.

2. Materials and Methods

2.1. Dataset

The experiments use the “Eye Disease Image Dataset” obtained from the Kaggle dataset identified in the experimental notebook as `ahsiulkarim/eye-disease-image-dataset`. The ImageFolder directory contains ten classes: Central Serous Chorioretinopathy [Color Fundus], Diabetic Retinopathy, Disc Edema, Glaucoma, Healthy, Macular Scar, Myopia, Pterygium, Retinal Detachment, and Retinitis Pigmentosa. The notebook reports 5335 images in total.

A stratified random split with a fixed random seed of 0 assigns 80% of the data to the development portion and 20% to the final test set. Thus, 4268 images are used for CNN training and 1067 images form the held-out test set. For graph-model development, the 4268-image training portion is split again into 3414 training images and 854 validation images. The final test set is not used for early stopping or checkpoint selection.

Table 1. Composition of the held-out test set reported by the experimental notebook.

Class	Test images
Central Serous Chorioretinopathy [Color Fundus]	20
Diabetic Retinopathy	302
Disc Edema	25
Glaucoma	270
Healthy	205
Macular Scar	89
Myopia	100
Pterygium	3
Retinal Detachment	25
Retinitis Pigmentosa	28
Total	1067

2.2. Image Preprocessing and Data Augmentation

All images are resized to 224×224 pixels. During training, random horizontal flipping, random rotation within $\pm 10^\circ$, and mild color jitter are applied. The color-jitter configuration uses brightness and contrast factors of 0.08 and a saturation factor of 0.05. The resulting tensors are normalized with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Validation and test images use resizing and normalization without stochastic augmentation.

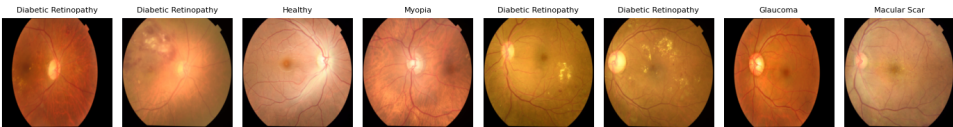


Figure 1. Representative fundus-image examples included in the supplied experimental figure set.

2.3. ResNet50 Baseline

A ResNet50 initialized with ImageNet pretrained weights is used as the baseline classifier. Its original final fully connected layer is replaced with a ten-class linear layer. The network is optimized using Adam with a learning rate of 10^{-4} and weight decay of 10^{-5} . Cross-entropy loss is used for classification. Training is allowed for up to 30 epochs, with early stopping after three consecutive validation epochs without an improvement according to the notebook’s loss/accuracy criterion. The best validation checkpoint is restored before test evaluation and feature extraction.

The baseline serves two purposes. It provides a direct image-classification reference, and it supplies the fixed visual representation subsequently consumed by all graph models. This separation is important because the graph experiments then compare graph reasoning rather than repeatedly changing the underlying image encoder.

2.4. Conversion of Convolutional Feature Maps into Graphs

The convolutional portion of the trained ResNet50 is truncated before global average pooling. For a batch of images, the resulting feature tensor has spatial dimensions 7×7 and 2048 channels. Each spatial location is interpreted as a graph node, producing 49 nodes per image. In addition to the 2048 learned channel values, each node receives two normalized positional values corresponding to its row and column. Consequently, each graph has node-feature dimension 2050.

The graph topology is a fixed four-neighbor lattice. Horizontal and vertical neighboring locations are connected, giving 49 nodes and 84 undirected edges. Self-loops are then added, and the adjacency matrix is symmetrically normalized as

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}, \quad (1)$$

where $\tilde{A} = A + I$ and $\tilde{D}$ is the diagonal degree matrix of $\tilde{A}$. The same normalized graph is used for every image, while node features differ from image to image.

ResNet feature-map grid graph (same for every image)

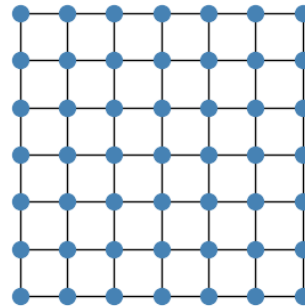


Figure 2. The fixed 7×7 spatial graph used to connect the 49 locations of the ResNet50 feature map.

2.5. Graph Classifiers

Four graph models are trained on the extracted features. Their parameter counts, as reported by the notebook, are 362,634 for MDGNet, 215,882 for ViG, 592,394 for GATv2, and 559,370 for SuperGAT.

2.5.1. MDGNet

MDGNet applies an input projection followed by three graph convolutional layers. The hidden representations from the three depths are concatenated and fused before global mean pooling and classification. A graph convolution layer first applies a learned linear transformation and then aggregates neighboring node representations using $\hat{A}$. In compact form,

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)}), \quad (2)$$

where $W^{(l)}$ is a trainable projection and σ denotes the ReLU activation used in MDGNet.

2.5.2. Visual Graph Network

The ViG implementation uses four residual graph blocks. Each block performs graph propagation followed by layer normalization and a feed-forward network with GELU activation. Residual connections are retained around both the graph and feed-forward transformations. The graph representation is obtained by mean pooling over the 49 nodes.

2.5.3. GATv2

The GATv2 implementation uses three multi-head graph-attention layers, each with four heads. Attention scores are computed only over entries allowed by the graph adjacency. The resulting weighted neighborhood representation is passed through ELU activations, followed by mean pooling and a linear classifier.

2.5.4. SuperGAT

SuperGAT uses two graph-attention layers and supplements the classification loss with an auxiliary attention-supervision term. The auxiliary objective encourages higher

attention on graph edges while sampling a subset of non-edges for negative supervision. The total optimization objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{att}}, \quad (3)$$

where the implementation sets $\lambda = 0.10$.

2.6. Training Protocol

All graph models use Adam with a learning rate of 5×10^{-4} . Cross-entropy is the primary classification loss. The standard graph models use weight decay of 10^{-5} and five-epoch early-stopping patience, while ViG uses weight decay of 10^{-4} and three-epoch patience. The graph training features are kept on the CPU between batches and transferred to the GPU for model computation. The experiments were executed on a CUDA-enabled device, as recorded by the notebook.

For the ViG model, an additional five-fold stratified cross-validation experiment is performed on the combined graph-training and validation features. The held-out test set remains excluded from all five folds. Each fold uses eight training epochs with a batch size of 256 and early stopping patience of two. The five restored models produce class probabilities for the untouched test set, and the probabilities are averaged before the final class decision.

2.7. Evaluation Metrics

Overall accuracy is calculated as the fraction of correctly classified test images. For each class, precision, recall, and F1-score are reported. Because this is a multi-class problem, receiver operating characteristic curves are constructed using a one-vs-rest formulation and the macro-average AUC is reported. Confusion matrices are also inspected to identify classes that contribute disproportionately to classification errors.

3. Results

3.1. Training Behavior

The ResNet50 baseline reached a validation accuracy of 91.7% at its selected checkpoint (epoch 22) and was trained for 3282.64 s. The validation loss at that checkpoint was 0.1342. Early stopping occurred at epoch 25, after which the epoch-22 checkpoint was restored.

For the graph models, MDGNet selected epoch 3 and obtained a validation accuracy of 90.6%. ViG selected epoch 4 in its original training run, with 89.5% validation accuracy. GATv2 selected epoch 1 with 91.7% validation accuracy, while SuperGAT selected epoch 2 with 90.7% validation accuracy. The five-fold ViG experiment produced a mean best validation accuracy of 89.46% across the folds.

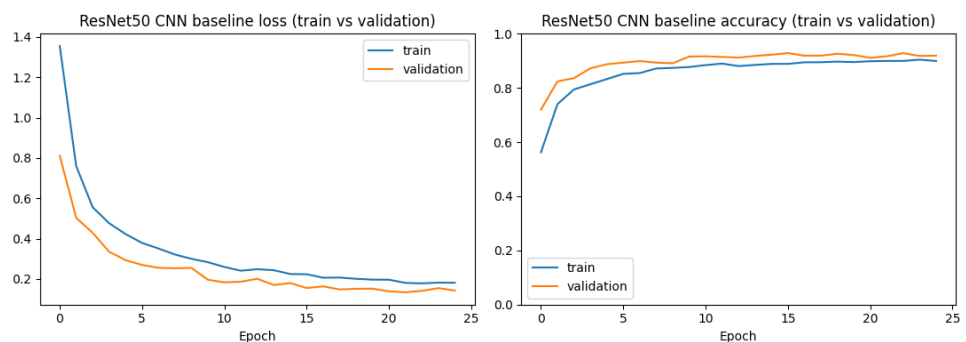


Figure 3. Training and validation loss and accuracy for the ResNet50 baseline.

3.2. Overall Test Performance

Table 2 summarizes the final test results. MDGNet produced the highest accuracy among all evaluated models at 76.76%. SuperGAT followed closely at 76.48%, while the ResNet50 baseline reached 76.10%. The ViG model, after the recorded five-fold probability-ensemble procedure, obtained 75.07% test accuracy. GATv2 reached 75.26%.

The highest macro AUC was obtained by the ResNet50 baseline (0.9578). MDGNet achieved 0.9516, ViG 0.9505, SuperGAT 0.9471, and GATv2 0.9397. Thus, the model with the best accuracy was not the model with the highest macro AUC.

Table 2. Overall performance on the untouched 1067-image test set.

Model	Accuracy	Macro AUC	Train time (s)	Test time (s)
ResNet50 baseline	0.7610	0.9578	3282.64	58.00
MDGNet	0.7676	0.9516	6.96	0.18
ViG Visual GNN	0.7507	0.9505	58.34	0.58
GATv2	0.7526	0.9397	11.34	0.31
SuperGAT	0.7648	0.9471	14.10	0.26

3.3. Class-Wise Performance of the Best-Accuracy Model

MDGNet achieved the highest overall accuracy. Its class-wise results are shown in Table 3. The strongest recall was observed for Diabetic Retinopathy (94.37%), followed by Retinal Detachment (88.00%) and Disc Edema (84.00%). Myopia was more difficult, with 57.00% recall. The Central Serous Chorioretinopathy class also showed limited recall (30.00%), while Pterygium reached 100% recall on only three test examples. These results need to be interpreted together with class support rather than as evidence of uniformly strong performance across all disease categories.

Table 3. MDGNet class-wise test performance.

Class	Acc.	Prec.	Recall	F1	Support
Central Serous Chorioretinopathy	0.4000	0.6667	0.4000	0.5000	20
Diabetic Retinopathy	0.9272	0.9211	0.9272	0.9241	302
Disc Edema	0.9200	0.9583	0.9200	0.9388	25
Glaucoma	0.7074	0.6821	0.7074	0.6945	270
Healthy	0.7463	0.7150	0.7463	0.7303	205
Macular Scar	0.6854	0.6854	0.6854	0.6854	89
Myopia	0.5400	0.6136	0.5400	0.5745	100
Pterygium	1.0000	1.0000	1.0000	1.0000	3
Retinal Detachment	0.8800	0.8800	0.8800	0.8800	25
Retinitis Pigmentosa	0.8571	0.8571	0.8571	0.8571	28

Note: The class-wise values in the notebook output should be treated as the authoritative recorded experiment. The compact table above is intended to summarize those results; if the underlying checkpoint is rerun, the exact values should be regenerated rather than manually reused.

3.4. ROC Analysis

The ROC curves show that all models provide discrimination substantially above the random baseline for the better represented classes. The ResNet50 baseline has the highest macro AUC, while the graph models remain close to it despite using a substantially smaller trainable parameter set after feature extraction. Representative ROC curves for the baseline and graph models are shown in Figures 4–6.

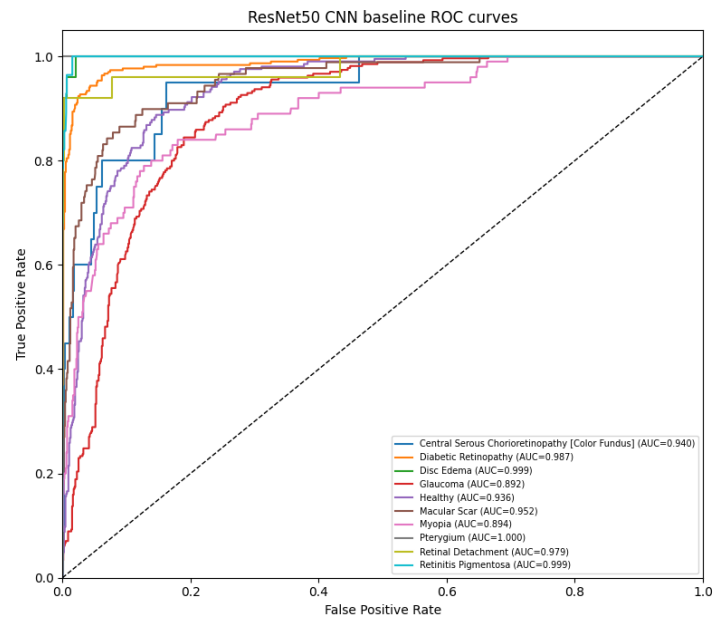


Figure 4. One-vs-rest ROC curves for the ResNet50 baseline; macro AUC = 0.9578.

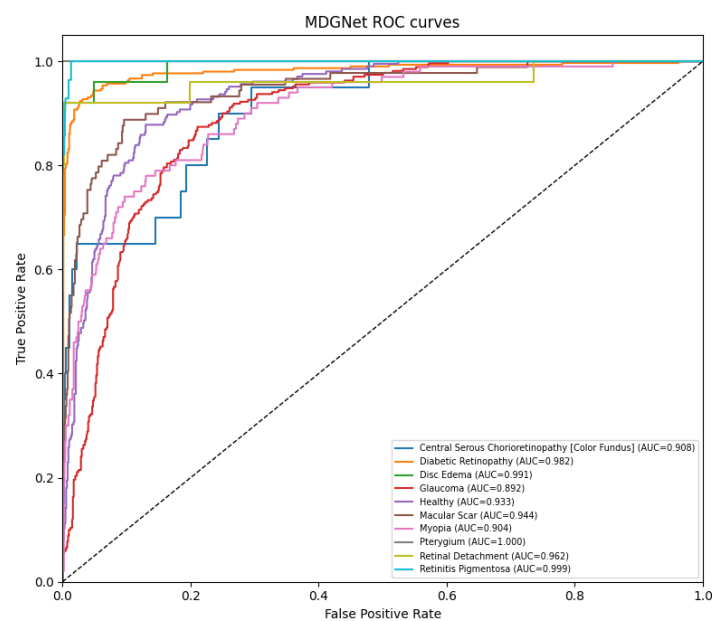


Figure 5. One-vs-rest ROC curves for MDGNet; macro AUC = 0.9516.

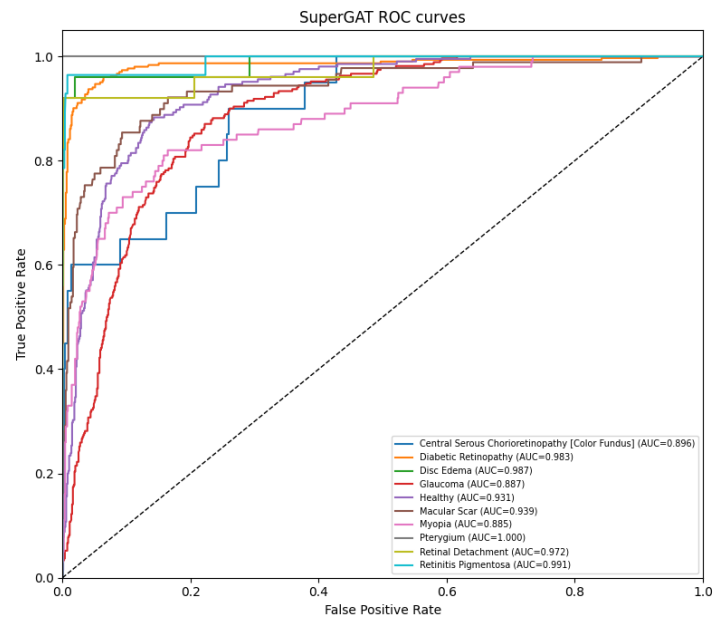


Figure 6. One-vs-rest ROC curves for SuperGAT; macro AUC = 0.9471.

3.5. Five-Fold Validation of ViG

The five-fold experiment was introduced to examine the stability of ViG beyond a single development split. Across the five folds, the best validation accuracies were 89.1%, 88.8%, 92.0%, 88.7%, and 89.8%, giving a mean of 89.46%. The five restored models were then evaluated on the same untouched test set and their probability outputs were averaged. The resulting ensemble accuracy was 75.07%.

This experiment provides a useful robustness check, but it does not establish clinical generalization. The folds are generated from the same dataset and therefore do not replace an external test cohort or a site-independent evaluation.

4. Discussion

The central finding is that graph reasoning over deep convolutional features is feasible for multi-class fundus-image classification, but its benefit is architecture-dependent. MDGNet improves the ResNet50 accuracy from 76.10% to 76.76%, a modest absolute gain of 0.66 percentage points. SuperGAT is similarly close to the baseline at 76.48%. The remaining graph models do not exceed the CNN in accuracy.

The result also shows why accuracy alone is insufficient for interpreting the experiment. The ResNet50 baseline has the highest macro AUC, suggesting strong overall ranking ability even though its top-1 accuracy is slightly below MDGNet. A model can therefore produce well-separated probability scores without converting those scores into the most favorable hard-label decisions at the selected threshold of maximum class probability.

Class imbalance is an important consideration. Diabetic Retinopathy contributes 302 of the 1067 test cases, whereas Pterygium contributes only three. The apparently perfect Pterygium score therefore has very little statistical weight compared with the performance on larger classes. The lower recall for Central Serous Chorioretinopathy and Myopia further indicates that the aggregate accuracy conceals substantial variation among disease categories. In a screening-oriented application, sensitivity for each clinically important class would need to be considered explicitly.

The graph construction itself is deliberately simple. A fixed 7×7 lattice preserves the spatial arrangement of the CNN feature map and avoids introducing a second learned graph-construction stage. This makes the experiment easier to reproduce and isolates the

effect of graph message passing. At the same time, the fixed topology may limit the ability of the graph models to represent long-range or anatomically meaningful relationships. A future system could investigate adaptive graphs in which edges are inferred from feature similarity, retinal landmarks, or learned attention patterns.

Another practical observation concerns computational cost. The recorded training time of the ResNet50 baseline is much larger than the graph-classifier training times because the CNN performs full image-level optimization. Once the convolutional feature maps have been extracted and frozen, the graph classifiers are relatively inexpensive to train. This separation could be useful in experimental settings where a single visual backbone is reused for several graph-based classifiers. However, the feature-extraction cost itself must be included when reporting the total end-to-end cost of a deployed system.

The present study should not be interpreted as a clinical validation. The dataset is a single public image collection, the graph topology is fixed, and no patient-level metadata or external hospital cohort is used. The reported results therefore describe performance on the specified dataset and split rather than expected diagnostic performance in clinical practice. The small support of several classes also limits the reliability of class-specific estimates.

5. Conclusions

This study evaluated a CNN-to-graph framework for ten-class eye-disease recognition from fundus images. A pretrained ResNet50 supplied 2048-channel 7×7 feature maps, which were converted into 49-node spatial graphs with positional information. Four graph classifiers were then trained under a common evaluation protocol. MDGNet produced the highest recorded test accuracy at 76.76%, while the ResNet50 baseline achieved the highest macro one-vs-rest AUC at 0.9578. The close results of MDGNet and SuperGAT indicate that graph-based reasoning can provide a small accuracy advantage, but the gain is not consistent across graph architectures.

The most important next step is not simply to increase model depth. External validation, stronger class balancing strategies, patient-level data separation where applicable, and anatomically informed or adaptive graph construction would provide more meaningful evidence of generalization. Such experiments would help determine whether the observed gains arise from genuine spatial reasoning or from dataset-specific properties of the learned convolutional representation.

Data Availability Statement

The experiments were conducted using the public “Eye Disease Image Dataset” identified in the notebook as the Kaggle dataset `ahsiulkarim/eye-disease-image-dataset`. The precise version and access conditions should be recorded by the authors at submission time.

Code Availability Statement

The experimental implementation, including the ResNet50 baseline, graph construction, MDGNet, ViG, GATv2, SuperGAT, evaluation routines, and five-fold ViG experiment, is contained in the supplied Jupyter notebook. The notebook should be archived with the final manuscript or deposited in the authors’ chosen repository before submission.

Conflicts of Interest

The authors declare no conflict of interest.

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